How to Write a Survey Paper: From Literature Search to Defensible Research Gaps

What is a survey paper?

- A survey paper reviews existing research on a specific topic to:
  - Understand what has been done.
  - Compare methods and findings.
  - Identify limitations, research gaps, and future directions.

- The goal is to understand the research field as a whole, not just summarize individual papers.

- A survey may be narrative, critical, systematic, or a combination of these approaches.

Types* of Survey Papers

Type of Survey	Purpose	Focus of Analysis	Typical Presentation
Thematic / Taxonomy	Classifying and organizing a research field	Categories, methods, and architectural similarities	Taxonomy or classification diagram
Evaluative / Benchmark	Comparing methods or tools	Performance measures, evaluation criteria, and trade-offs	Comparison tables and graphs
Application-Oriented	Examining applications in a particular domain	Practical effectiveness and applicability	Pipeline diagrams
Evolutionary	Understanding the development of a field	Milestones, trends, and paradigm shifts	Timeline or trend analysis

* These approaches may overlap; a survey can combine more than one.A single survey may

Traditional / Critical Survey
vs.
Systematic Literature Review

Feature	Traditional / Critical Survey	Systematic Literature Review
Primary goal	Understand the field, **compare approaches**, and identify challenges	Answer **specific research questions systematically**
Search strategy	May be **selective**; documentation practices vary	**Documents databases, search terms, and selection procedures**
Paper selection	Guided by **expert judgment and relevance**	Uses **predefined inclusion** and **exclusion criteria**
Analysis and synthesis	**Critically compares** methods, themes, strengths, and limitations	**Synthesizes evidence across selected studies** to answer research questions
Quality assessment	Formal assessment **may be used**, depending on the review	May use **predefined criteria** to assess study quality, where appropriate
Transparency and reproducibility	Depends on how clearly the review process is **documented**	**Explicitly reports review procedures** to support transparency and reproducibility

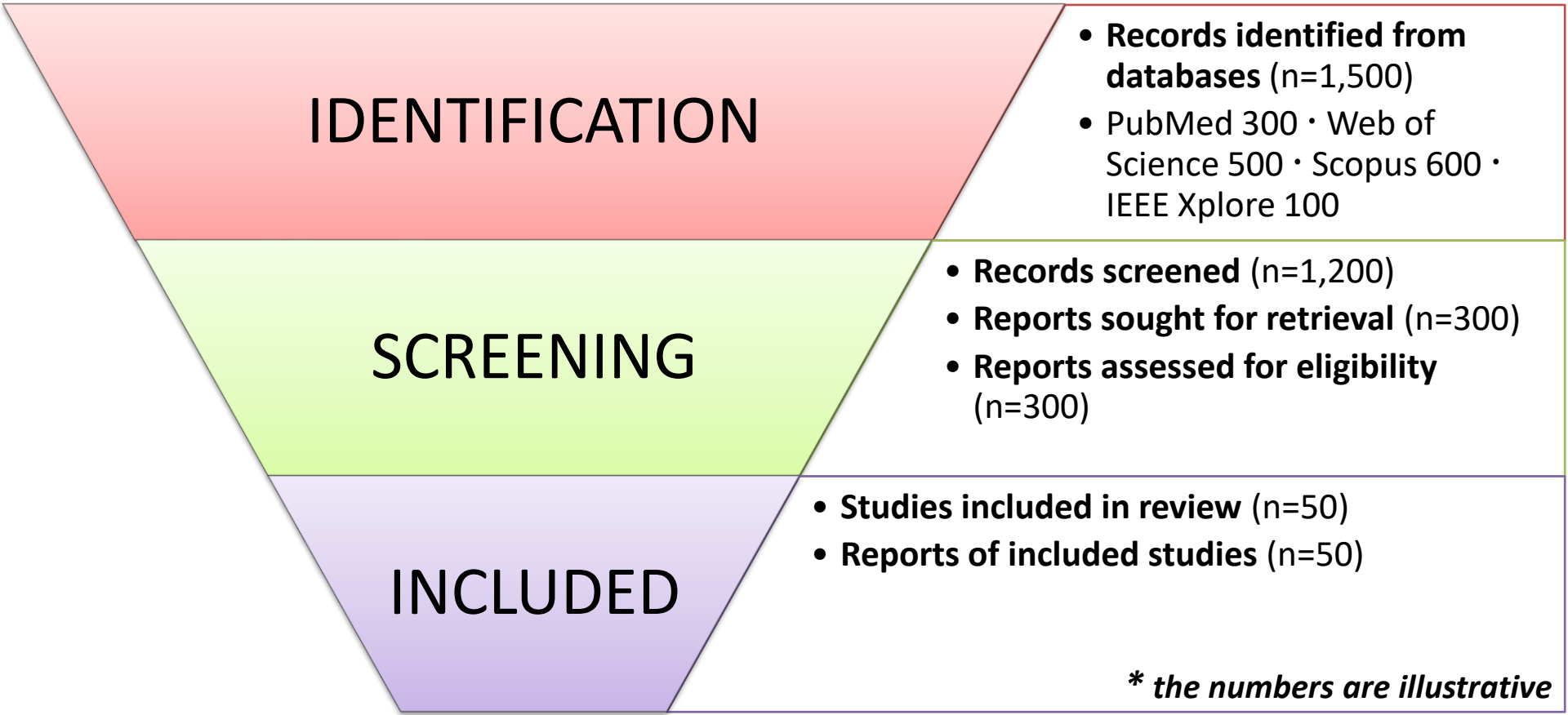
PRISMA 2020

- **P**referred **R**eporting **I**tems for **S**ystematic Reviews and **M**eta-**A**nalyses.
- A widely used **reporting guideline** for systematic reviews.
- Promotes **transparent, complete, and clear reporting** of the review process.
- **PRISMA guides reporting, not how to conduct the research.**

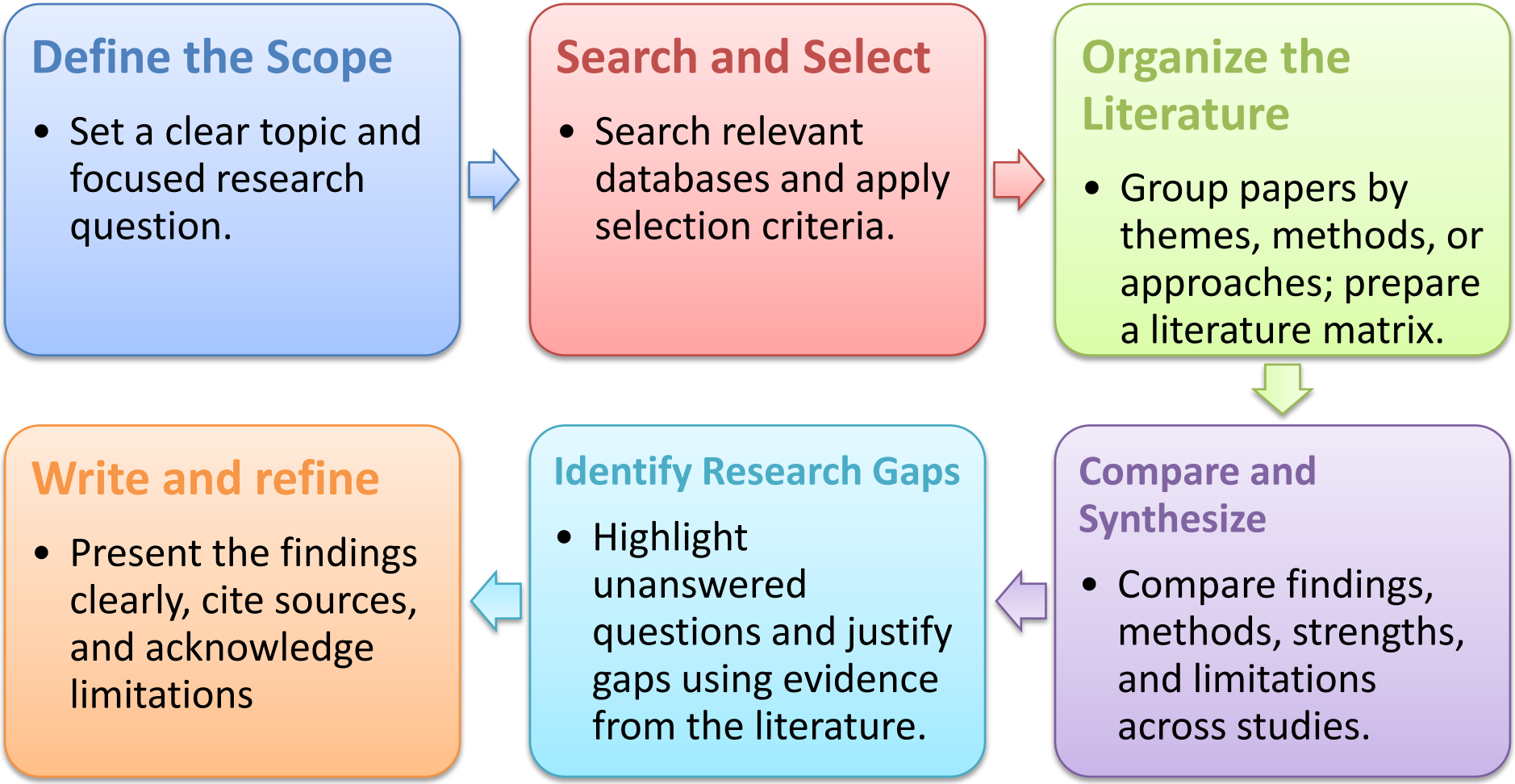
https://www.prisma-statement.org/prisma-2020

Key Components of PRISMA 2020

- 27-Item Checklist
- A **3-step funnel** that visually maps out a literature search:



Practical Steps for Writing a Survey Paper



Do's and Don'ts

- ✓ **Do look for patterns, not just papers.**
  - Connect findings to reveal broader insights.
- ✓ **Do question the evidence.**
  - Examine assumptions, data, and metrics.
- ✓ **Do identify gaps through synthesis.**
  - Uncover contradictions and unresolved questions.
- ✓ **Do offer a perspective.**
  - Develop a useful framework or interpretation.

- × **Don't make the scope too broad.**
  - Keep the focus manageable.
- × **Don't just summarize papers.**
  - Synthesize findings and compare evidence.
- × **Don't assume research gaps.**
  - Validate them against existing literature.
- × **Don't prioritize quantity, novelty, or AI over quality.**
  - Focus on meaningful insights and verified evidence.

Stay curious. Keep questioning.

Happy Research!